
Mathematical Background

This preliminary handout collects the notation, definitions and basic facts that the lectures assume without proof. Everything specific to linear programming — convex sets and polyhedra, duality, the simplex method, interior point methods — is developed from scratch beginning in Lecture 1; what follows is background only. Each item carries a gray annotation indicating where in the course it is first needed. Proofs of all stated facts can be found in any standard text on linear algebra, calculus and probability.

1 Linear Algebra

Notation. We write $\mathbb{Z}$ for the set of integers, $\mathbb{R}$ for the reals, and $[n] := \{1, \dots, n\}$. Vectors $x \in \mathbb{R}^n$ are column vectors with coordinates $x_1, \dots, x_n$; the standard basis vectors are $e_1, \dots, e_n$, and $\mathbf{0}$ and $\mathbf{1}$ denote the all-zeros and all-ones vectors. Inequalities between vectors are coordinatewise: $x \leq y$ means $x_i \leq y_i$ for all i . A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ has entries a_{ij} and transpose $\mathbf{A}^\top$. For index sets $I \subseteq [m]$ and $J \subseteq [n]$, the submatrix $\mathbf{A}_{I,\bullet}$ consists of the rows of $\mathbf{A}$ indexed by I , and $\mathbf{A}_{\bullet,J}$ of the columns indexed by J ; for single indices, $\mathbf{A}_{i,\bullet}$ is the i -th row and $\mathbf{A}_{\bullet,j}$ the j -th column. When a section uses only one kind of indexing, we lighten the notation by declaring locally that $\mathbf{A}_I$ abbreviates $\mathbf{A}_{I,\bullet}$ (or $\mathbf{A}_{\bullet,I}$). Subvectors are unambiguous: x_I denotes the subvector of coordinates of x indexed by I . (*this indexing is used constantly, beginning with tight constraint sets in Lecture 1*)

DEFINITION 1 (SPAN, INDEPENDENCE, BASIS, DIMENSION) A (linear) subspace $W \subseteq \mathbb{R}^n$ is a set closed under addition and scalar multiplication. The span of $v^1, \dots, v^k \in \mathbb{R}^n$ is the subspace $\text{span}(v^1, \dots, v^k) := \{\sum_{i=1}^k \lambda_i v^i : \lambda \in \mathbb{R}^k\}$ of all their linear combinations. The vectors are linearly independent if $\sum_{i=1}^k \lambda_i v^i = \mathbf{0}$ forces $\lambda = \mathbf{0}$. A basis of W is a linearly independent set spanning W ; all bases of W have the same size, called the dimension $\dim(W)$.

DEFINITION 2 (RANK) The rank of $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the dimension of the span of its columns, which always equals the dimension of the span of its rows: $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$. The matrix has full column rank if $\text{rank}(\mathbf{A}) = n$ and full row rank if $\text{rank}(\mathbf{A}) = m$.

THEOREM 3 (LINEAR SYSTEMS) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, and define the kernel and image

$$\ker(\mathbf{A}) := \{x \in \mathbb{R}^n : \mathbf{A}x = \mathbf{0}\}, \quad \text{im}(\mathbf{A}) := \{\mathbf{A}x : x \in \mathbb{R}^n\}.$$

- (a) The system $\mathbf{A}x = b$ is solvable if and only if $b \in \text{im}(\mathbf{A})$.
- (b) If x^0 is one solution, the set of all solutions is the translate $x^0 + \ker(\mathbf{A})$.
- (c) $\dim \ker(\mathbf{A}) = n - \text{rank}(\mathbf{A})$ (rank-nullity).
- (d) The solution is unique if and only if $\mathbf{A}$ has full column rank. In particular, a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ with $\text{rank}(\mathbf{A}) = n$ is invertible: there is a unique matrix $\mathbf{A}^{-1}$ with $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$, and $x = \mathbf{A}^{-1}b$ solves $\mathbf{A}x = b$.

(vertices arise as unique solutions of tight subsystems in Lecture 1; bases drive the simplex method in Lecture 2; each interior point iteration in Lecture 5 solves one linear system)

Inner product and norm. The inner product of $x, y \in \mathbb{R}^n$ is $x^\top y = \sum_{i=1}^n x_i y_i$, and the Euclidean norm is $\|x\| := \sqrt{x^\top x}$. Vectors with $x^\top y = 0$ are orthogonal.

THEOREM 4 (CAUCHY-SCHWARZ) For all $x, y \in \mathbb{R}^n$, $|x^\top y| \leq \|x\| \|y\|$, with equality if and only if x and y are parallel. Consequently the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$ holds. (proximity to the central path is measured in this norm throughout Lectures 4–5)

DEFINITION 5 (ORTHOGONAL COMPLEMENT) The orthogonal complement of a subspace $W \subseteq \mathbb{R}^n$ is the subspace $W^\perp := \{y \in \mathbb{R}^n : y^\top x = 0 \text{ for all } x \in W\}$.

THEOREM 6 (ORTHOGONAL DECOMPOSITION) For any $A \in \mathbb{R}^{m \times n}$,

$$\ker(A)^\perp = \text{im}(A^\top),$$

and every $x \in \mathbb{R}^n$ decomposes uniquely as $x = u + A^\top v$ with $u \in \ker(A)$ and $A^\top v \in \text{im}(A^\top)$ orthogonal to each other. In particular, a vector orthogonal to all of $\ker(A)$ is of the form $A^\top y$ for some $y \in \mathbb{R}^m$.

REMARK 1 Theorem 6 is the single most used identity in the second half of the course: it converts the geometric condition “ g is orthogonal to every direction in $\ker(A)$ ” into the existence of multipliers y with $g = A^\top y$. This is the source of the dual variables accompanying the central path in Lectures 4–5, and it reappears below in the characterization of log-barrier minimizers.

DEFINITION 7 (AFFINE SUBSPACES, HULLS, AND DIMENSION) An affine subspace of $\mathbb{R}^n$ is a translate $A = x^0 + W$ of a linear subspace W ; its dimension is $\dim(W)$. The affine hull $\text{aff}(S)$ of a set $S \subseteq \mathbb{R}^n$ is the smallest affine subspace containing it, equivalently the set of affine combinations $\sum_{i=1}^k \lambda_i x^i$ with $x^i \in S$ and $\sum_{i=1}^k \lambda_i = 1$. The dimension of an arbitrary set S is that of its affine hull. A hyperplane is an affine subspace of dimension $n - 1$, equivalently a set $\{x : a^\top x = \beta\}$ with $a \neq 0$; it bounds the halfspace $\{x : a^\top x \leq \beta\}$. (the dimension of faces of polyhedra, from Lecture 1 onward)

2 One-Variable Calculus

Remarkably little calculus is needed in this course: the interior point lectures use only the following facts about functions of a single real variable, applied to restrictions of multivariate functions to line segments.

FACT 8 (LOGARITHM) For $t > 0$: $\frac{d}{dt} \ln t = \frac{1}{t}$ and $\frac{d^2}{dt^2} \ln t = -\frac{1}{t^2}$.

THEOREM 9 (FERMAT’S RULE) If a differentiable function $\varphi : (a, b) \rightarrow \mathbb{R}$ attains a local minimum at $t_0 \in (a, b)$, then $\varphi'(t_0) = 0$.

DEFINITION 10 (CONVEXITY) A function $\varphi : I \rightarrow \mathbb{R}$ on an interval $I \subseteq \mathbb{R}$ is convex if

$$\varphi(\lambda s + (1 - \lambda)t) \leq \lambda \varphi(s) + (1 - \lambda)\varphi(t) \quad \text{for all } s, t \in I, \lambda \in [0, 1],$$

and strictly convex if the inequality is strict whenever $s \neq t$ and $\lambda \in (0, 1)$.

THEOREM 11 (SECOND-DERIVATIVE TEST) A twice differentiable $\varphi : I \rightarrow \mathbb{R}$ is convex if and only if $\varphi'' \geq 0$ on I ; if $\varphi'' > 0$ on I , then φ is strictly convex.

THEOREM 12 (STATIONARY POINTS OF CONVEX FUNCTIONS) *Let $\varphi : I \rightarrow \mathbb{R}$ be convex and differentiable. If $\varphi'(t_0) = 0$, then t_0 is a global minimizer of φ on I . If φ is strictly convex, it has at most one minimizer.*

DEFINITION 13 (GRADIENT) *For $f : U \rightarrow \mathbb{R}$ with $U \subseteq \mathbb{R}^n$ open and partial derivatives existing and continuous, the gradient at $x \in U$ is $\nabla f(x) := (\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x))$. Along the line through x in direction $d \in \mathbb{R}^n$,*

$$\frac{d}{dt}f(x + td) = \nabla f(x + td)^\top d.$$

REMARK 2 (WHAT THIS IS FOR) These facts are combined in Lecture 4 to characterize the minimizer of a log-barrier problem

$$\min c^\top x - \sum_{i=1}^n \ln x_i \text{ s.t. } \mathbf{A}x = b, \quad x > \mathbf{0}.$$

Along a feasible line $x + td$ with $d \in \ker(\mathbf{A}) \setminus \{\mathbf{0}\}$, the objective restricts to a one-variable function with second derivative $\sum_{i=1}^n d_i^2 / (x_i + td_i)^2 > 0$, hence strictly convex. Fermat's rule (Theorem 9), Theorem 12 and the orthogonal decomposition (Theorem 6) then show: if the problem has a minimizer, it is unique, and a feasible $x > \mathbf{0}$ is that minimizer exactly when $\nabla f(x) \perp \ker(\mathbf{A})$, equivalently $\nabla f(x) = \mathbf{A}^\top y$ for some $y \in \mathbb{R}^m$.

3 Asymptotic Notation, Computation, and Graphs

DEFINITION 14 (ASYMPTOTIC NOTATION) *For $f, g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$, we write $f(n) = O(g(n))$ if there are constants $C, n_0 > 0$ with $f(n) \leq C g(n)$ for all $n \geq n_0$; $f(n) = \Omega(g(n))$ if $g(n) = O(f(n))$; and $f(n) = \Theta(g(n))$ if both hold. (all complexity statements, from Lecture 2 onward)*

DEFINITION 15 (BIT SIZE) *A rational number p/q in lowest terms, with $p, q \in \mathbb{Z}$ and $q \geq 1$, is encoded in binary using $O(\log(|p| + 1) + \log(q + 1))$ bits. The encoding length L of an LP instance with rational data $(\mathbf{A}, b, c)$ is the total number of bits needed to write down all its entries. (the complexity landscape of Lecture 2 and the iteration bounds of Lecture 4)*

FACT 16 (COST OF LINEAR ALGEBRA) *A system of n linear equations in n unknowns can be solved — or its unsolvability detected — by Gaussian elimination using $O(n^3)$ arithmetic operations. (the per-iteration cost of the simplex and interior point methods)*

DEFINITION 17 (GRAPHS) *An (undirected) graph $G = (V, E)$ consists of a finite vertex set V and a set E of unordered pairs of vertices, the edges. A path between u and v is a sequence of distinct vertices starting at u and ending at v in which consecutive vertices are joined by edges; its length is its number of edges. The graph is connected if every pair of vertices is joined by a path; the distance $d_G(u, v)$ is the minimum length of a u - v path, and the diameter of a connected graph is $\max_{u, v \in V} d_G(u, v)$. (the vertex-edge graph of a polyhedron and its diameter, Lectures 2–3)*

4 Probability

Probability enters the course only in Lecture 3, on the smoothed analysis of the simplex method. We work at an informal level with continuous random variables described by densities.

DEFINITION 18 (DENSITY) A random variable X has density $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ if $\int_{\mathbb{R}} f(t) dt = 1$ and

$$\Pr[X \in S] = \int_S f(t) dt \quad \text{for all (measurable) } S \subseteq \mathbb{R}.$$

A random vector $X \in \mathbb{R}^n$ with joint density $f : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ is defined analogously, with integrals over subsets of $\mathbb{R}^n$.

DEFINITION 19 (UNIFORM RANDOM VARIABLE) X is uniform on the interval $[a, b]$ if it has density $f(t) = \frac{1}{b-a}$ for $t \in [a, b]$ and $f(t) = 0$ otherwise: all subintervals of equal length are equally likely.

DEFINITION 20 (EXPECTATION) The expectation of a random variable X with density f is $\mathbb{E}[X] := \int_{\mathbb{R}} t f(t) dt$, and more generally $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(t) f(t) dt$.

THEOREM 21 (LINEARITY OF EXPECTATION) For any random variables $X_1, \dots, X_k$ and scalars $a_1, \dots, a_k$,

$$\mathbb{E}\left[\sum_{i=1}^k a_i X_i\right] = \sum_{i=1}^k a_i \mathbb{E}[X_i],$$

regardless of any dependence between the X_i .

DEFINITION 22 (INDICATOR OF AN EVENT) The indicator of an event A is the random variable $\mathbb{1}_A$ that equals 1 if A occurs and 0 otherwise.

FACT 23 (EXPECTATIONS OF INDICATORS) $\mathbb{E}[\mathbb{1}_A] = \Pr[A]$: the expectation of an indicator is the probability of its event. Combined with linearity of expectation, this converts counting into probability: the expected number of the events $A_1, \dots, A_k$ that occur is $\mathbb{E}[\sum_{i=1}^k \mathbb{1}_{A_i}] = \sum_{i=1}^k \Pr[A_i]$. (used constantly in Lecture 3)

DEFINITION 24 (INDEPENDENCE AND CONDITIONAL DENSITIES) Random vectors X and Y are independent if their joint density is the product of their individual densities: $f(x, y) = f_X(x) f_Y(y)$; equivalently, $\Pr[X \in S, Y \in T] = \Pr[X \in S] \cdot \Pr[Y \in T]$ for all S, T . In general, the conditional density of X given $Y = y$ is

$$f(x | y) := \frac{f(x, y)}{f_Y(y)}, \quad \text{where } f_Y(y) := \int f(x, y) dx > 0$$

is the marginal density of Y . Expectations satisfy the tower rule $\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}[X | Y]]$. (the smoothed analysis of Lecture 3 conditions on partial information about the random constraints)

THEOREM 25 (UNION BOUND) For any events $A_1, \dots, A_k$,

$$\Pr\left[\bigcup_{i=1}^k A_i\right] \leq \sum_{i=1}^k \Pr[A_i].$$

This is all the probability — and, together with the preceding sections, all the background — that the course requires.